



Course name: INTERMEDIATE MACROECONOMICS
Course code: EC2211
Type of exam: Retake
Examiner: Martin Flodén
Number of credits: 7.5
Date of exam: Thursday 4 December 2025
Examination time: 5 hours (08:00-13:00)
Aids: No aids are allowed

Write your anonymization code on each answer sheet (the code you received from Ladok when you registered for the exam).

Start each new question on a new answer sheet.

Explain notions/concepts and symbols. If you think that a question is vaguely formulated, specify the conditions used for solving it. Only legible exams will be marked.

The exam consists of 5 questions, worth 100 points in total. The maximum number of points on each sub-question are stated within parenthesis. For the grade E 45 points are required, for D 50 points, C 60 points, B 75 points and A 90 points.

All questions should be answered in English.

Your results will be made available on your Ladok account (www.student.ladok.se) within 15 working days from the date of the examination.

Good luck!



Question 1: Short questions (30 points in total)

Please write no more than one page (maximum) when answering each of these questions.

- a. Consider an economy where the production function is

$$Y_t = A_t K_t^\alpha H_t^\beta L_t^{1-\alpha-\beta}$$

where A is TFP, K is physical capital, H is human capital and L is labor.

Suppose $\alpha = \beta = 1/4$ and that the growth rates of output, the capital stock, human capital and labor are

$$g_Y = 3\%, g_K = 4\%, g_H = 1\% \text{ and } g_L = 1\%.$$

Calculate the TFP growth rate. (5 points)

- b. Consider the production function

$$Y_t = F(K_t, L_t) = AK_t^\alpha + AL_t^{1-\alpha}$$

where $0 < \alpha < 1$.

Does this production function display constant returns to scale? Explain. (5 points)

- c. Explain the Permanent Income Hypothesis. What key assumptions does the theory rely on? (5 points)
- d. An individual lives one period, works H hours, and earns no income besides labor income. Her utility function is

$$U(C, H) = \alpha \ln(C) + (1 - \alpha) \ln(1 - H)$$

where $0 < \alpha < 1$. The wage rate is w .

Write down the budget constraint and solve for optimal C and H . (5 points)

- e. In the years after World War II, when much of Europe lay in ruins, the United States ran a trade surplus. Use your knowledge about how the trade balance relates to international capital flows to provide a hypothesis that could explain what was going on. (5 points)
- f. State the Fisher equation. Explain what is meant by the Taylor principle and the logic behind that principle. (5 points)



Question 2: The Solow model (25 points)

Consider the Solow model. Your task is to use the model to analyze the consequences of a fall in the savings rate. A fraction s of income is invested in capital every period, and capital depreciates at rate δ .

Production is Cobb-Douglas:

$$Y_t = AK_t^\alpha L_t^{1-\alpha}$$

where TFP, A , is constant, and $0 < \alpha < 1$. The population, L , grows at the constant rate n .

- Derive an expression for the accumulation of capital per person in this economy, i.e., Δk_{t+1} , where $k_t \equiv K_t/L_t$. (4 points)
- State the steady state condition for capital per person. Illustrate the steady state in a diagram. (4 points)
- Suppose the economy starts in steady state. Analyze the effects of an immediate and permanent decrease in the savings rate.

Use the model to explain what happens to the economy over time and in the long run. In particular, comment on what happens to the levels and growth rates of per capita GDP and per capita consumption. (17 points)

Question 3: Policy spillover to a small open economy (25 points)

Consider a small open economy with a floating exchange rate. The economy starts with GDP at potential, its interest rate at the natural level and inflation at the inflation target. Then a recession abroad leads the foreign central bank to temporarily cut its policy rate.

- Suppose the Home central bank keeps its own policy rate unchanged. Use the interest-parity condition to analyze how the exchange rate responds. (5 points)
- Suppose that developments abroad affect the Home economy only through the exchange rate via interest parity.

Use the AS-AD framework to analyze how the economy is affected over time. Explain how the output gap, the nominal interest rate, inflation and inflation expectations develop over time. (15 points)

- In part (b), we assumed that the only spillover occurred through interest parity.

Suggest one additional effect that the foreign recession might have on the Home economy, and explain briefly how this would affect the analysis in part (b). (5 points)

Question 4: Debt sustainability and price stability (10 points)

This is an essay question where you are supposed to refer to theories and material covered in the course but you need not use equations or graphs in your answer. Write no more than 2 pages (maximum).

Public debt has increased and concerns have arisen about the sustainability of public finances.

Explain how concerns about fiscal sustainability can threaten price stability, and discuss the trade-offs the central bank may face in this situation.

When answering, assume the country has a flexible exchange rate.



Question 5: The time consistency problem (10 points)

This is a credit question. Students with credit from the exercises will be awarded the better of that credit and the result on this question. I suggest that you only answer this question if you lack sufficient credit from the exercises.

Consider a one-period economy.

At the start of the periods, households form inflation expectations π^e . After inflation expectations are formed, the policymaker chooses the actual inflation rate π .

Unemployment is given by the Phillips curve:

$$u = \bar{u} - \phi(\pi - \pi^e).$$

The policymaker chooses π to minimize:

$$L(\pi, y) = \pi^2 + \lambda u,$$

where λ measures the weight the policymaker places on low unemployment.

- a. Solve for the policymaker's optimal choice of inflation π as a function of π^e , $\bar{u}$ and other parameters. (4 points)
- b. Households understand the policymaker's incentives and minimization problem and anticipate her behavior. Therefore, $\pi^e = \pi$.

What is the equilibrium inflation rate and unemployment rate? (4 points)

- c. Explain briefly why the parameter λ affects the equilibrium outcome. (2 points)